

Why STE-Top-K Pruning Fails: a derivation

Base network: a frozen MLP 784-1024-1024-10, two hidden layers of $N_1 = N_2 = 1024$ neurons ($N = 2048$). The goal is to choose a binary keep-mask. We build the argument from the gate definition up to the structural reason top-K loses to the soft penalty.

1. The selection problem

A pruner assigns each hidden neuron j a score s_j and a gate g_j in $\{0,1\}$. The pruned network keeps neuron j iff $g_j = 1$. We want the mask of a fixed size K that hurts accuracy least:

$$\min_{m \in \{0,1\}^N} \mathcal{L}(\theta \odot m) \quad \text{subject to} \quad \sum_j m_j = K.$$

This is combinatorial. Both methods relax it: turn scores into gates differentiably and descend. They differ only in HOW the gate is formed -- and that difference is everything.

2. The straight-through estimator and where its gradient lives

A hard gate is an indicator with a threshold t ,

$$g_j = 1[s_j > t],$$

whose derivative is zero almost everywhere. The straight-through estimator keeps the hard value in the forward pass but substitutes a sigmoid in the backward pass:

$$g_j = \text{sg}(1[s_j > t] - \sigma(s_j)) + \sigma(s_j),$$

where sg is stop-gradient. The forward value equals the hard gate; the backward derivative is

$$\frac{\partial g_j}{\partial s_j} = \sigma'(s_j) = \sigma(s_j)(1 - \sigma(s_j)),$$

so by the chain rule the learning signal that reaches score s_j is

$$\frac{\partial \mathcal{L}}{\partial s_j} = \frac{\partial \mathcal{L}}{\partial g_j} \sigma'(s_j).$$

The factor $\sigma'(s)$ is the entire story. It is a bump centred at 0 that decays exponentially:

$$\sigma'(s) = \frac{e^{-|s|}}{(1 + e^{-|s|})^2} \leq e^{-|s|}, \quad \max_s \sigma'(s) = \sigma'(0) = \frac{1}{4}.$$

So a neuron only learns if its score is near 0. Where is the keep/kill boundary? It is the threshold t . The signal is useful only when the boundary coincides with the bump, $t = 0$.

Soft penalty (lambda). The threshold is fixed at $t = 0$, and a +2.0 bias plus the penalty hold scores near 0. Boundary = bump centre: every contested neuron sits where sigma' is large. Top-K. The threshold is the K-th largest score, an order statistic

$$\tau = s_{(K)},$$

which is generally far from 0. The neurons that decide the mask (those near tau) sit on the exponential tail of sigma', so the boundary signal is

$$\sigma'(\tau) \leq e^{-|\tau|} \approx 0 \quad \text{when } |\tau| \text{ is large.}$$

Consequence: plain STE freezes the ranking. One-shot top-K at 24% kept = 42.7pp (lambda ~2pp).

3. Fix 1: centre the surrogate on the moving threshold

Re-place the bump onto tau with a temperature T:

$$\text{soft}_j = \sigma\left(\frac{s_j - \tau}{T}\right), \quad \frac{\partial g_j}{\partial s_j} = \frac{1}{T} \sigma'\left(\frac{s_j - \tau}{T}\right).$$

Now the bump is centred at $s = \tau$ and has width $\sim T$: a neuron learns iff $|s_j - \tau| \leq cT$. This restores lambda's free property (gradient on the boundary).

Fixes the easy regime: 50% kept 13.9 -> 2.1pp.

But cold-start at aggressive K still collapses: 24% kept -> 88.3pp (one class).

The width-T band only works if scores live on a scale near T, which forces a normalisation of the scores. That normalisation is where the two real pathologies enter.

4. Pathology 1: standardisation + global top-K starves a layer

Standardise each layer's raw scores a to zero mean and unit variance, then add a per-layer context bias c (the only cross-layer degree of freedom):

$$z_j^{(\ell)} = \frac{a_j^{(\ell)} - \mu_\ell}{\sigma_\ell} \Rightarrow \text{mean}_j z_j^{(\ell)} = 0, \quad \text{var}_j z_j^{(\ell)} = 1,$$

$$s_j^{(\ell)} = z_j^{(\ell)} + c_\ell.$$

Global top-K keeps the K largest scores across BOTH layers. Survivors in layer l:

$$n_\ell = |\{j: s_j^{(\ell)} \geq \tau\}| = |\{j: z_j^{(\ell)} \geq \tau - c_\ell\}|.$$

Model the standardised scores as Gaussian, $z \sim N(0,1)$, with Phi the standard normal CDF. Then

$$E[n_\ell] = N_\ell (1 - \Phi(\tau - c_\ell)),$$

and the budget fixes tau implicitly through

$$\sum_{\ell} N_{\ell} (1 - \Phi(\tau - c_{\ell})) = K.$$

Because both layers are centred and unit-variance, the ONLY thing that tilts the split is the bias gap. For the two equal layers ($N_1 = N_2 = N$), write $\Delta = c_2 - c_1$:

$$\frac{n_1}{n_2} = \frac{1 - \Phi(\tau - c_1)}{1 - \Phi(\tau - c_2)}.$$

Now take Δ large at fixed K . Layer 2 alone can supply the whole budget, so tau locks to layer 2: $\tau = c_2 + \Phi^{-1}(1 - K/N)$. Then the argument for layer 1 is

$$\tau - c_1 = \Delta + \Phi^{-1}\left(1 - \frac{K}{N}\right) \rightarrow +\infty,$$

$$\Rightarrow n_1 = N(1 - \Phi(\tau - c_1)) \rightarrow 0.$$

So an unbounded bias gap drains layer 1 entirely. And Δ IS unbounded: we removed the tanh, c is the only free cross-layer knob, and its gradient does not vanish, so SGD lets it drift.

Observed: layer-1 survivors 819 -> 512 -> 205 -> 0. At $K = N_1$ the first layer is severed.

A severed first layer makes every logit constant in the input, so the network predicts the majority class:

$$\text{acc} \rightarrow \max_c \pi_c \approx 0.11 \Rightarrow \text{drop} \approx 88 \text{ pp.}$$

Fix: bound the bias. $c = \tanh(\hat{c})$ in $(-1,1) \Rightarrow \Delta$ in $(-2,2)$. With unit-variance z the two clouds always overlap; worst case $\Delta = 2$ at $\tau = \text{mean}$ gives

$$\frac{n_1}{N} = 1 - \Phi(1) \approx 0.159, \quad \frac{n_2}{N} = 1 - \Phi(-1) \approx 0.841 \text{ (both } > 0).$$

5. Pathology 2: the standardisation freezes the encoder

Look at the normalisation as a function of the raw encoder output a (per layer):

$$y(a) = \frac{a - \mu(a)}{\sigma(a)}.$$

Mean and std are homogeneous of degree 1, so for any scale $\alpha > 0$,

$$\mu(\alpha a) = \alpha \mu(a), \quad \sigma(\alpha a) = \alpha \sigma(a) \Rightarrow y(\alpha a) = y(a).$$

Thus the loss is invariant to the encoder's output scale: it is homogeneous of degree 0,

$$\mathcal{L}(\alpha a) = \mathcal{L}(a) \quad \forall \alpha > 0.$$

By Euler's theorem for a degree-0 homogeneous function, the radial derivative vanishes:

$$a \cdot \nabla_a \mathcal{L} = 0.$$

So the scale $\|a\|$ is a flat direction of the loss -- unconstrained, it random-walks upward under SGD noise. Worse, with a detached std the gradient that does reach a is

$$y = \frac{a - \mu}{\sigma_{\text{det}}} \Rightarrow \frac{\partial \mathcal{L}}{\partial a} = \frac{1}{\sigma} \frac{\partial \mathcal{L}}{\partial y} \Rightarrow \left\| \frac{\partial \mathcal{L}}{\partial a} \right\| \propto \frac{1}{\sigma}.$$

As sigma drifts up the encoder gradient vanishes and the ranking ossifies -- fine for easy pruning, useless for precise high-sparsity choices.

Observed: raw encoder std 0.2 -> 148; the per-neuron ranking stops updating.

Fix: centre-only, $y(a) = a - \mu(a)$. This is degree-1, NOT degree-0:

$$y(\alpha a) = \alpha y(a) \neq y(a) \Rightarrow \mathcal{L}(\alpha a) \neq \mathcal{L}(a),$$

so the scale is loss-constrained (no free growth) and the gradient carries no $1/\sigma$ factor ($d\mathcal{L}/da = d\mathcal{L}/dy$). The encoder keeps learning; std plateaus ~ 21 and aggressive-K drops halve.

Aside: weight decay also bounds sigma, but its per-step shrinkage is $\eta * \lambda_{\text{wd}}$. With $\eta = 1e-3$, $\lambda_{\text{wd}} = 1e-2$ gives $1e-5$ (100x too weak, sigma -> 141); $\lambda_{\text{wd}} = 1$ bounds sigma but over-regularises the whole pruner (15pp). Centre-only removes the cause instead.

6. The residual, and why top-K loses even when stable

With both fixes there is no collapse, but a structural gap remains. The per-step signal to neuron j is

$$\left| \frac{\partial \mathcal{L}}{\partial s_j} \right| = \left| \frac{\partial \mathcal{L}}{\partial g_j} \right| \cdot \frac{1}{T} \sigma' \left(\frac{s_j - \tau}{T} \right),$$

non-negligible only on the band $|s_j - \tau| \leq cT$. If the scores have spread Σ , the fraction of neurons that learn each step is

$$\text{active fraction} \approx \frac{cT}{\Sigma}.$$

Measured at the target sparsity (24% kept): only $\sim 7.7\%$ of neurons receive a gradient per step.

Contrast the soft penalty. Its sparsity term contributes, to EVERY neuron simultaneously,

$$\frac{\partial}{\partial s_j} \left(\lambda \frac{1}{N} \sum_i \sigma(s_i) \right) = \frac{\lambda}{N} \sigma'(s_j),$$

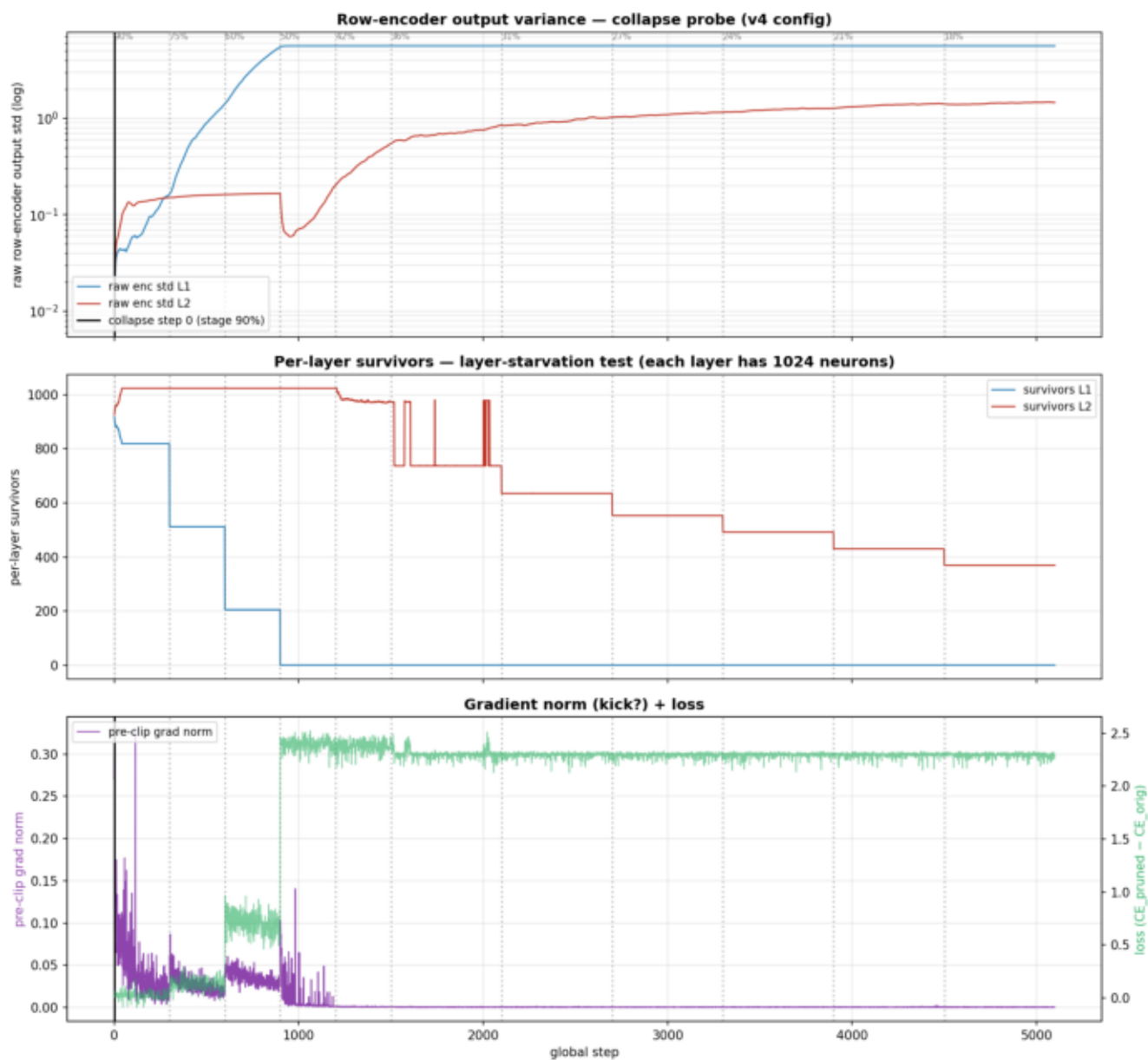
and with scores held near 0 (the +2.0 bias) every neuron sits near the σ' peak. So lambda delivers a dense, GLOBAL gradient over the whole set, while top-K delivers a sparse, boundary-LOCAL one. Optimising a combinatorial subset by local boundary exchanges is weaker than joint global descent -- hence the $\sim 2.6x$ gap, with no numerical blow-up to blame.

7. Verdict

Fully de-bugged top-K (centered STE + centre-only + tanh, 2 seeds, no collapse):
34.6% pruned at 2pp; 5.29 +/- 0.36pp at lambda's 76%-pruned operating point.
Soft penalty (lambda): 76% pruned at 2pp (~2.0pp there). Top-K prunes ~2.6x less.

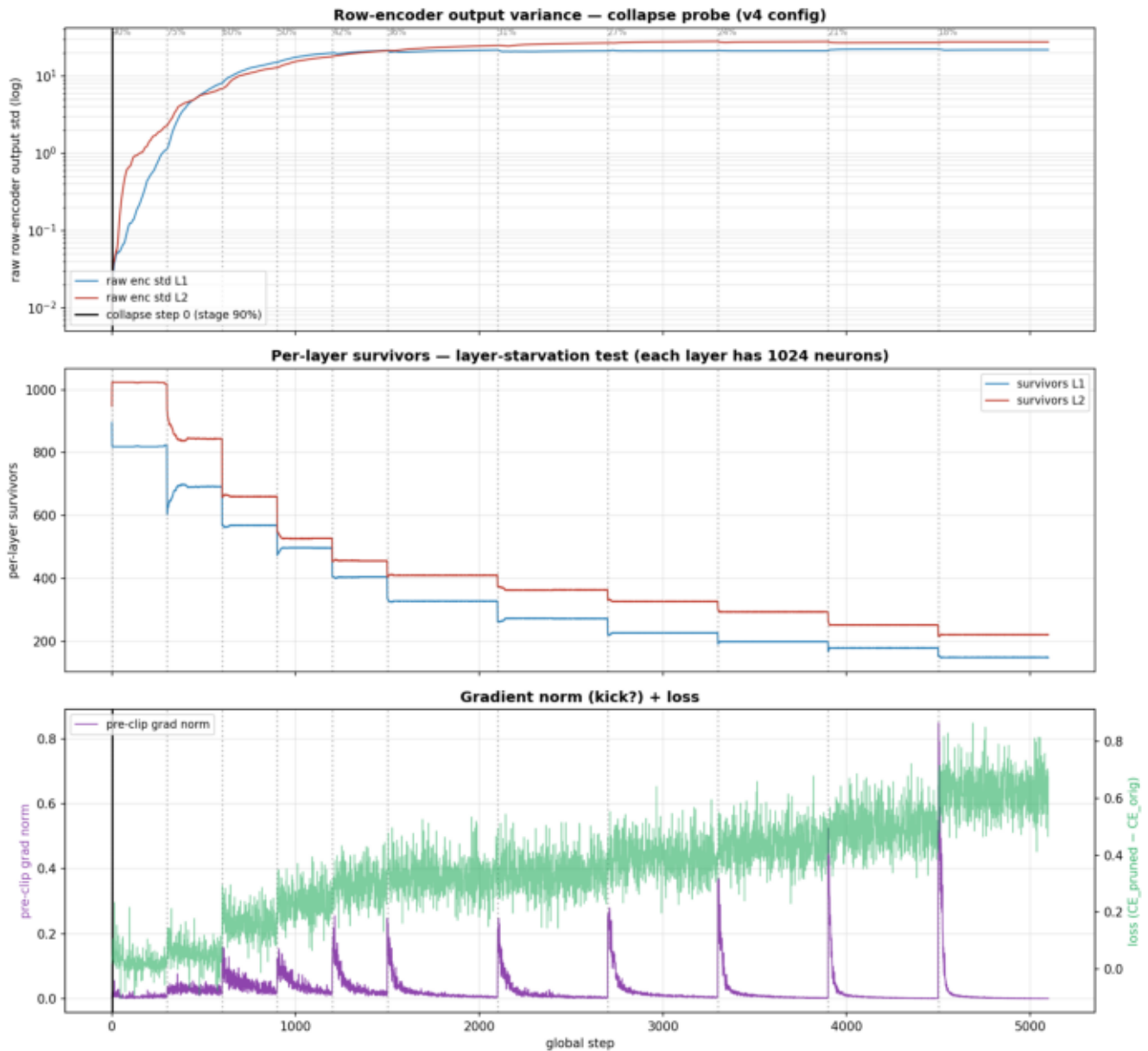
Three problems, each derived above: (1) plain STE saturates because the bump is at 0 but the boundary is tau; (2) standardisation + unbounded bias starves a layer; (3) standardisation is scale-invariant and freezes the encoder. Tanh and centre-only remove (2) and (3); the residual is the boundary-local vs global-gradient gap, which is intrinsic to a hard budget. The soft penalty needed none of these guardrails. Sweep-free is not the same as better.

Figure 1 -- Pathology 1: layer starvation



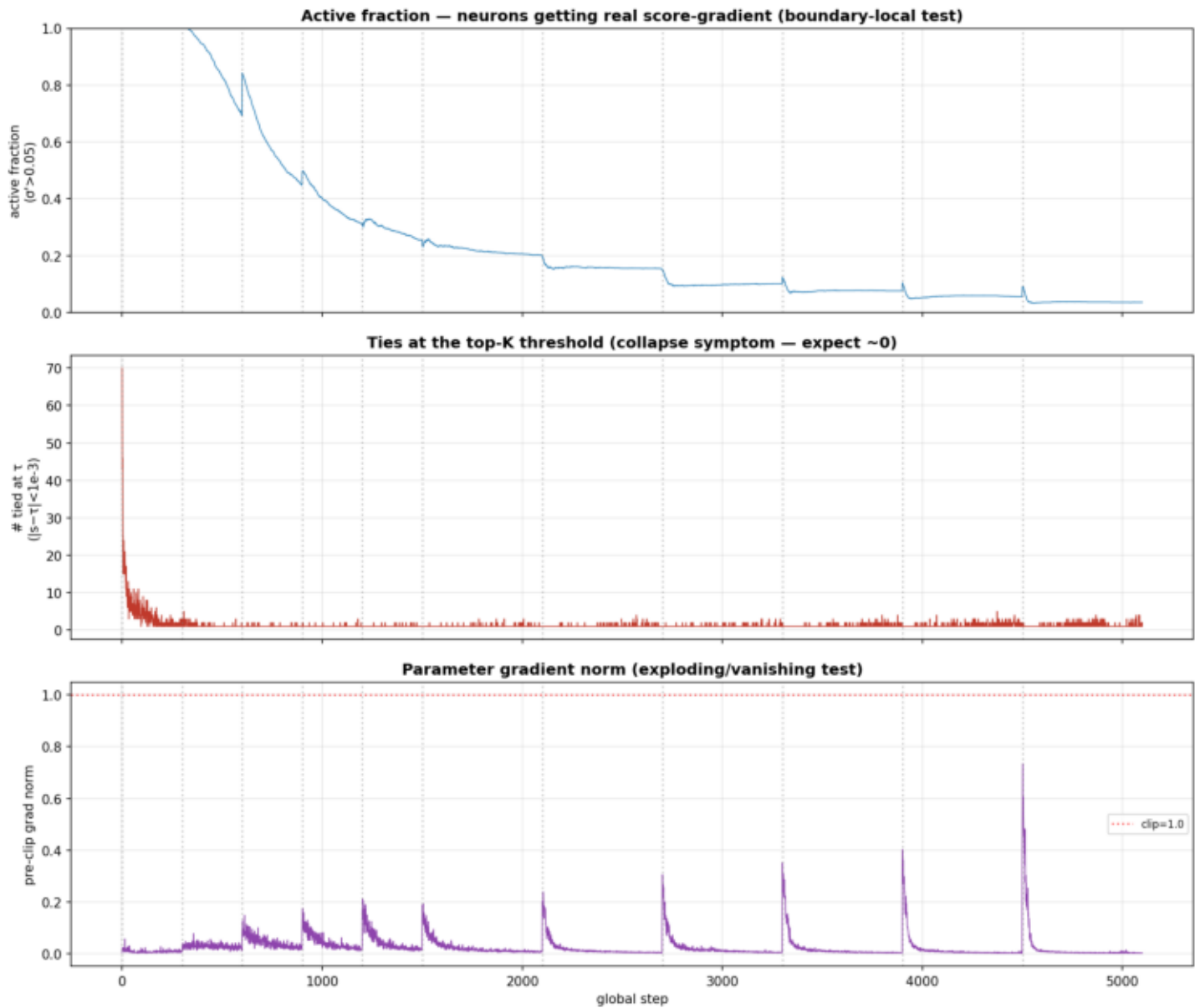
Middle: layer-1 survivors (blue) drain 819->512->205->0 while layer-2 (red) stays full; at K = one layer the first layer is severed and the network outputs one class (88.32pp).

Figure 2 -- Pathology 2 removed by centre-only



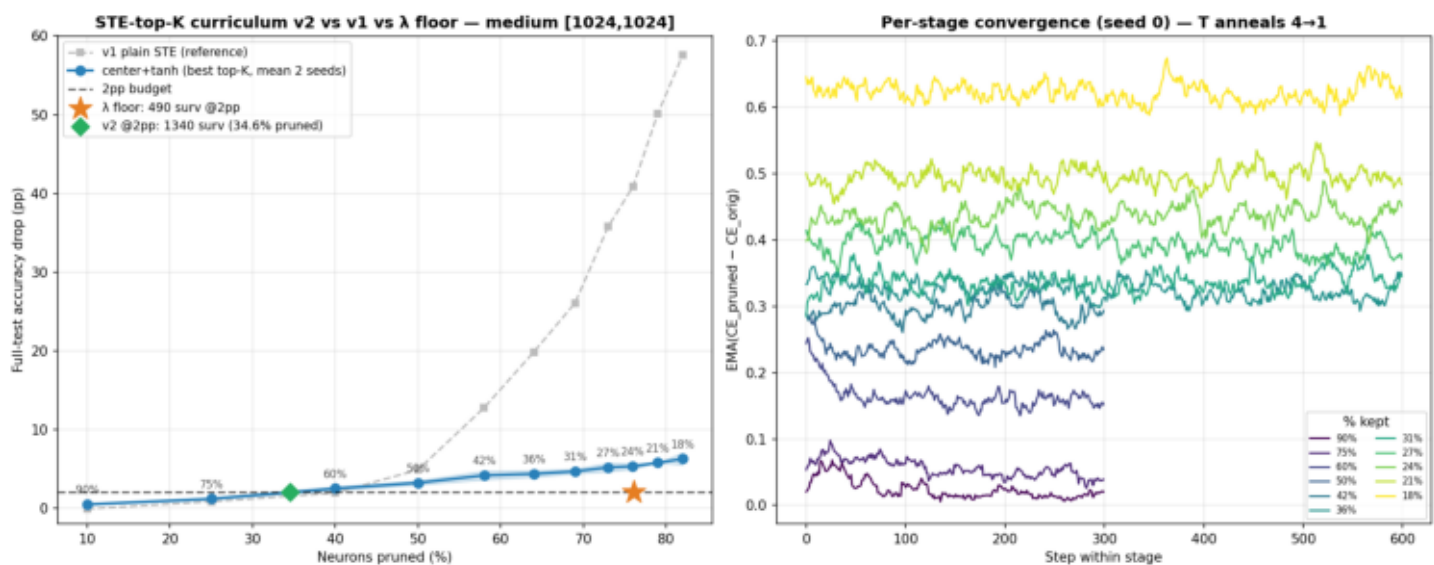
Top: raw encoder std now plateaus (~21) instead of running to 148. Bottom: the loss runs a per-stage sawtooth -- the encoder relearns each stage, i.e. it is no longer frozen.

Figure 3 -- The boundary-local gradient (structural reason)



Active fraction (neurons with a real score-gradient) falls toward ~ 0.08 at the target sparsity; ties ~ 0 and the parameter gradient is stable -- the deficit is locality, not numerics.

Figure 4 -- Even fully de-bugged, top-K loses to lambda



Best stable top-K (center+tanh, 2 seeds) crosses 2pp at ~35% pruned; the lambda floor (orange star) is at 76% pruned -- about 2.6x more pruning at the same accuracy.